

React State, Props & Events

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Objectives

- Props
- State
- Events in React

Props

- Props is short for properties
- Used to pass data between Components
- Similar to method parameters in Java and C#
- Arguments in JS
- Attributes in HTML

```
const getLastName = (name) => {  
  let lastName = name.split(' ')[1];  
  return <span>{lastName}</span>  
}  
export const NameList = (props) => (  
  <ul>  
    {props.names.map(name => (  
      <li key={name}>  
        <div>{name} - {name.toUpperCase()} -  
        {getLastName(name)}</div>  
      </li>  
    ))}  
  </ul>  
);
```

```
import ReactDOM from "react-dom";  
import { NameList } from "./NameList";  
  
const list = ["Mehrdad Javan", "Ulf Bengtsson ", "Simon Elbrink", "Erik Svensson"];  
  
ReactDOM.render(<NameList names={list} />, document.getElementById('root'));
```

State

- State is a plain JavaScript object that stores the dynamic data of a component
- As an object holds information that may change over the lifetime of the component
- Used to represent information about the component (current state)

```
import React from "react";
export default class Car extends React.Component {
  state = { brand: "Volvo", model: "V60" };

  render() {
    return (
      <div>
        <b>{this.state.brand} - {this.state.model}</b>
      </div>
    );
  }
}
```

```
componentDidMount() {
  let _volvoV60 = {...this.state};
  _volvoV60.model = "V90";
  this.setState(_volvoV60);
  console.log("data: ", _volvoV60);
}
```

componentDidMount is called immediately after the component is rendered

useState

- `useState()` is a new function that allows to have **state variables** in **function** components
- Returns a pair value: current state & a function that updates the current state
- **syntax:** `const [state, setState] = useState(initialstate);`

```
import React, { useState, useEffect } from "react";
```

```
export const Car = () => {  
  const initData = { brand: "Volvo", model: "V60" };  
  const [car, setCar] = useState(initData);
```

```
  return (  
    <div>  
      <b>  
        {car.brand} - {car.model} - {car.color}  
      </b>  
    </div>  
  );  
};
```

```
useEffect(() => {  
  setCar({...car, model: "V90", color: "Gray"});  
}, []);
```

`useEffect` is called immediately after the component is rendered

State vs Props

- State is an updatable structure that has information about the component
- Props is an object which stores the value of attributes of a tag

Props	State
Props are read-only.	State can be changed using events or lifecycle method
Props are immutable.	State is mutable.
props can be passed to component from outside	State can be passed to child components from inside a component
Props can be accessed by the child component.	State cannot be accessed by child components.
Props make components reusable.	State cannot make components reusable.

Event - *onClick*

- Is an action that could be triggered as a result of the user action or system generated event
- React has the same events as HTML: *click, change, mouseover ...*
- Events in React are written in **camelCase**

```
import React, { useState } from "react";
export const Counter = () => {
  const [counter, setCounter] = new useState(0);

  const incrementHandler = () => {
    setCounter(counter + 1);
  };
  return (
    <div>
      <h3>Counter: {counter}</h3>
      <button onClick={incrementHandler}>Increment</button>
      <button onClick={() => setCounter(counter - 1)}>Decrement</button>
    </div>
  );
};
```

Counter: 2

Increment

Decrement

Reset

Event - *onChange*

- *onChange* event detects when the value of an input element changes

```
import React, { useState } from "react";
export const Person = () => {
  const defaultData = { firstName: '', lastName: '' }
  const [person, setPerson] = useState(defaultData);

  return (
    <div>
      <form>
        <input type="text" id="firstName" name="firstName"
          value={person.firstName} onChange={(e) => onInputChange(e, 'firstName')} /><br/>
        <input type="text" id="lastName" name="lastName"
          value={person.lastName} onChange={(e) => onInputChange(e, 'lastName')} /><br/>
        <button type="button" onClick={submitPerson}>Submit</button>
      </form>
    </div>);
};
```

```
const onInputChange = (e, attributeName) => {
  const val = e.target.value;
  let _person = {...person};
  _person[`${attributeName}`] = val;
  setPerson(_person);
}

const submitPerson = () => {
  console.log(person);
}
```


Links

<https://www.javatpoint.com/react-state-vs-props>

<https://reactjs.org/docs/events.html>